# Non-linear fracture mechanics analysis of ground-supported slabs

**Ground-supported slabs represent one of the main applications of steel fibre-reinforced concrete (SFRC). In these structures fibres can totally substitute the conventional reinforcement (rebars or welded fabric) with significant advantages in terms of toughness and strength under static and dynamic loads. Furthermore, fibres can reduce cracking due to thermal or shrinkage effects.**

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**Acknowledgements:**

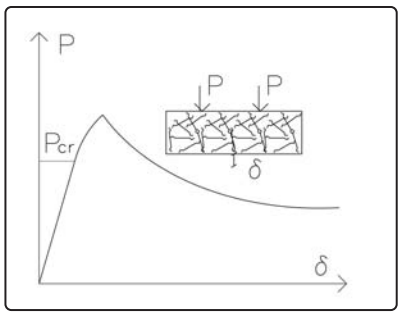
This research project was financed by Officine Maccaferri S.p.A. (Bologna, Italy) whose support is kindly acknowledged. The authors would also like to thank Professor Alberto Meda, Dr Luca Sorelli, Paolo Martinelli and Luca Cominoli for their collaboration in carrying out the experiments.

The use of fibre reinforcement is often economically beneficial with respect to conventional reinforcement due to reduction of labour costs and time to check the correct placement of the reinforcement. At present there is still a lack of design rules for SFRC structures. As a consequence, engineers usually design ground-supported slabs by adopting the same rules that are valid for concrete with conventional reinforcement and are based on the elastic behaviour of the slab. However, this approach is not correct since the post-cracking behaviour of SFRC is significantly different from concrete with conventional reinforcement. In fact, the post-cracking behaviour of SFRC is a markedly non-linear behaviour and softens with the volume fraction of fibres often used in practice ($V_f$=0.3-0.6%; see Figure 1a); the conventional reinforcement retains an (almost) linear behaviour (with a lower stiffness with respect to the uncracked stage) until the reinforcement yields (see Figure 1b).

## Design criteria

Therefore, it is clear that SFRC slabs require new or improved design rules which acknowledge the efficiency of FRC. A few published results are available, such as the well-known slabs tested by Falkner *et al.*[1] Other interesting experimental results were obtained by Beckett[2] and Kukreja[3], while earlier experimental results from the authors were published by Meda *et al.*[4–6].

In this article results will be presented from a new approach for analysing ground-supported slabs, based on non-linear fracture mechanics (NLFM). This approach is now adopted in the Italian design guidelines[7]. Numerical results based on NLFM are compared with results from an extensive experimental research programme on FRC slabs on an elastic subgrade.



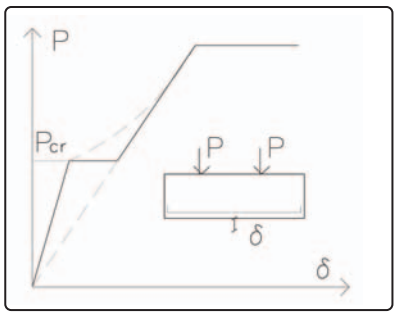




**Figure 1a below left: Typical response of a bending test on a steel-fibre-reinforced beam.**

**Figure 1b below right: Typical response of a bending test on a conventional reinforced concrete beam.**

## Experimental programme

The behaviour of slabs was experimentally studied on full-scale specimens: numerous SFRC slabs having a square geometry with a side of 3 and a thickness of 0.15m were tested[4–6]. The experimental model aims to reproduce a portion of slab between joints. A reference slab (P0) was cast without any reinforcement and slabs reinforced with steel fabric at the bottom were also tested.

Two different types of hooked end steel wire fibres have been adopted in the SFRC slabs (see Figure 2):

- 50/1.0 fibres having a length ($l_f$) of 50mm and a diameter ($\phi_f$) of 1mm (aspect ratio= $l_f/\phi_f$ =50);
- 30/0.6 fibres having a length ($l_f$) of 30mm and a diameter ($\phi_f$) of 0.6mm (aspect ratio=50).

The slabs were made of a normal strength concrete (C25/30) typical for ground-supported slab applications. The slump of the fresh concrete was higher than 150mm for all the slabs.

In order to reproduce an elastic (Winkler) soil, the slabs were placed on small steel springs (see Figure 3a) that were designed to simulate a typical sub-grade for pavements (k=0.08N/mm$^3$). The springs were introduced via small steel plates supported along the border (see Figure 3c); each spring was tested in order to verify the design stiffness. The target subgrade stiffness was obtained by placing the springs at a distance of 375mm in both directions. A thin layer of high-strength mortar was placed between the springs and the slab to compensate the out-of plane deformations of the slab due to curling (see Figure 3d).

The load was applied to the centre of the slab by means of an hydraulic jack (see Figure 3b). A 1000kN load cell was placed under the jack and six linear variable differential transformers (LVDTs) were used to measure the vertical displacements in different locations of the slab (see Figure 3b).

The fracture properties of the materials have been determined by four point bending tests on notched beams (150 × 150 × 600mm), according to the Italian Standard[8]. The bending tests were carried out using a closed-loop hydraulic machine (Instron) with the crack opening displacement as feedback control. The slabs were tested up to failure; the ultimate load was conventionally reached when two main cracks developed up to the slab border and a collapse mechanism occurred (see Figure 4). Afterwards, the load could still increase because of the subgrade stiffness.

The slab with no reinforcement not only has a lower ultimate load but also exhibits more brittle behaviour (the slab fell into four pieces at ultimate load).

## NLFM analyses

Since fibres start working after cracking of the concrete matrix and the material response is no longer linear, SFRC slabs on grade can be better analysed by adopting methods based on NLFM[9]. This approach pemits simulation of the crack development in the slab that is stable because of the stress redistribution along the cracked surfaces. The

**Figure 2: Steel fibre adopted for casting the slabs.**

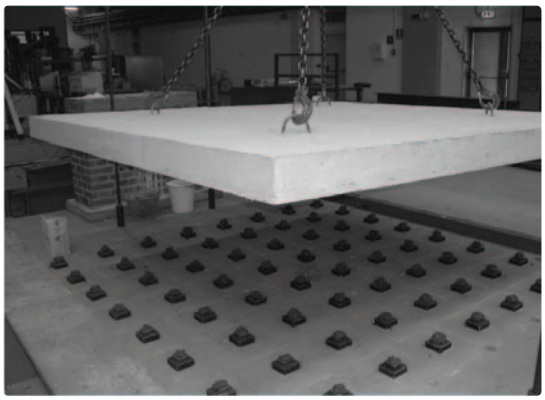
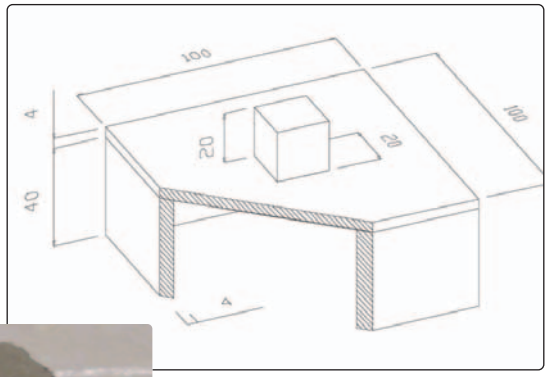


**Figure 3a right: Spring position simulating the Winkler soil.**

**Figure 3b far right: Experimental set-up.**



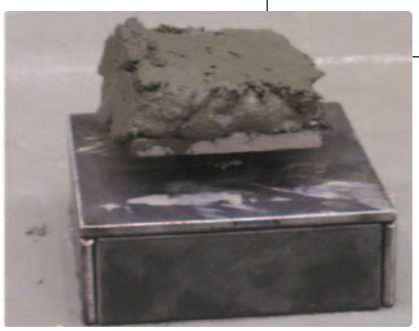



**Figure 3c and 3d: Steel spring for simulating the elastic subgrade.**

load increases during crack propagation until a collapse mechanism occurs; the ultimate load may be four to five times the first crack load[4]. Thus, the residual strength of cracked concrete, which is significant for SFRC, is taken into account in the constitutive laws.

The numerical analyses of SFRC ground-supported slabs can be performed by adopting Finite-Element Analysis (FEA) commercial programs that include NLFM based on a discrete crack or a smeared crack approach. The FEA program adopted for this article, Merlin[10], uses discrete cracks and considers the structure as a number of linear elastic sub-domains linked by interface elements. The latter simulate those cracks for which positions must be known *a priori*. Interface elements initially connect the sub-domains (as rigid links) and start activating (i.e. cracks start opening) when the normal tensile stress at the interface reaches the tensile strength of the material. Afterwards, the crack propagates and cohesive stresses are transmitted between the crack faces according to stress-crack opening (σ–w) laws (which are given in the input for the interface elements). These (σ–w) laws were obtained by performing an inverse analysis[11] from the fracture tests on beam specimens.

A typical comparison between numerical (NLFM) and experimental results, as obtained from a SFRC slab, is shown in Figure 5 where the numerical crack development is also shown. It can be seen that the numerical NLFM analyses provide a good approximation of the experimental behaviour, both for the initial stiffness and for the ultimate load. The numerical crack development is also shown in Figure 5. It should be noted that the similar final crack pattern and the significant load increment in the SFRC slab after crack onset was made possible by the remarkable toughness of SFRC and by the stress redistribution over the cracks.

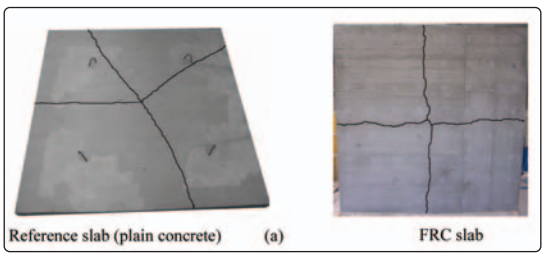



**Figure 4: Typical final crack patterns of ground-supported slabs tested with a point load in the slab centre.**

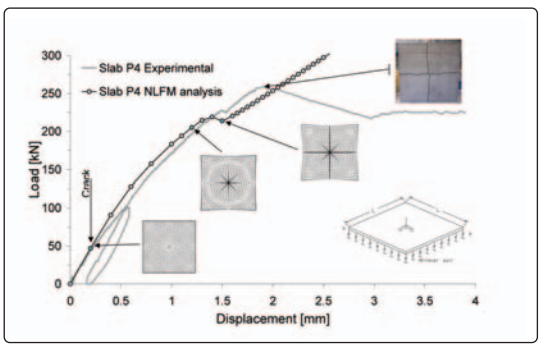



**Figure 5: Comparison of numerical and experimental results for a typical FRC slab[6].**

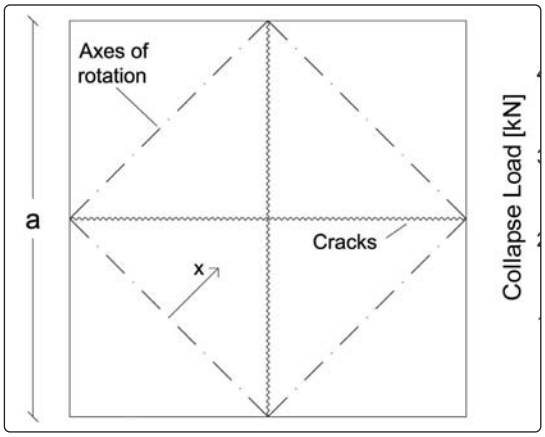



**Figure 6a: Yield lines in a ground-supported slab with free borders.**

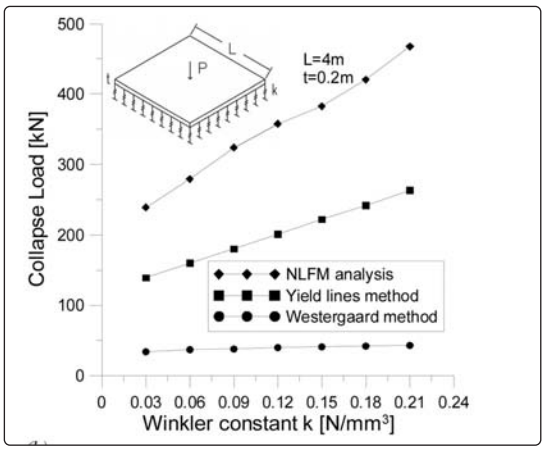



**Figure 6b: Comparison between maximum loads determined with different approaches[14].**

### Comparison with the yield line method

The remarkable fracture energy available in FRC can be accounted for in the evaluation of the ultimate load of FRC structures by using the yield line method[12,13]. However, since the energy for this limit analysis is generally a part of the total energy, approaches based on the yield line theory provide results on the safe side since they underestimate the actual ultimate load of the ground-supported slab[14]. A typical example is shown in Figure 6 where, for different values of the subgrade modulus, the ultimate load, determined with Westergaard, yield lines theory and NLFM theory, is compared.

The different value of the ultimate load can be explained by considering that the NLFM analyses used the entire fracture energy of FRC while the yield line approach was based on the residual tensile strength of FRC corresponding to large crack openings.

### Concluding remarks

This article presents an NLFM approach for analysing ground-supported slabs reinforced with a small volume fraction of steel fibres. Results show that steel fibres prevent the brittle collapse of slabs without reinforcement and provide remarkable slab strength.

Structural analyses of the slabs based on NLFM allow better simulation of the actual slab behaviour; by using this method, the load increase during the crack development can be accounted for until a collapse mechanism occurs. The NLFM approach has been validated by comparing the numerical and the experimental results. Comparisons with the yield line method are also presented. ■

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CURRENT PRACTICE SHEET

# Admixture current practice – *Part 1*

**This Current Practice Sheet (CPS) provides background information on current admixture use. Part 2 will look in more detail at admixtures, what they are, how they work and how they are used.**

**John Dransfield**, Cement Admixtures Association

Admixture sales have grown steadily over the past 15 to 20 years and current statistics suggest that over 75% of ready-mixed, site-mixed and precast concrete now contains an admixture. This makes admixtures a normal constituent of concrete but possibly not one where the benefits are being fully used. Total sales have more than doubled over this period and show that superplasticisers and 'other' special-purpose admixtures have become more important but that normal plasticisers remain the dominant admixture type. This split between superplasticisers and plasticisers is much lower than in most other Europe countries and probably reflects the fact that the UK still tends to specify a 50 or 75mm slump concrete which, by modern standards, is unnecessarily unworkable and difficult to place. In 2004, the split was 36% superplasticiser in the UK against a European average of 56%, with countries like Sweden, Switzerland and Italy running at over 90%.

### Applications

Admixture types and their applications are well documented and The Concrete Society Technical Report 18[1], revised in 2002, remains a good reference source. The main areas of application can be summarised as:

- economic – optimised mix design
- placing – appropriate consistence, compactability, cohesion, setting and strength development
- problem solving – concrete that will meet special needs for placing or performance
- durability – helping to ensure that concrete fulfils its design requirements for its intended life
- environmental – helping to reduce the health and safety aspects of concrete during placing and environmental impact during its life cycle.

A single admixture can often fulfil several of the above functions if used at an appropriate dose. However, there are situations that require two or more admixtures and UK suppliers provide an excellent technical advice service to help meet specific needs and optimise admixture use and performance.

The continued preference for many concrete purchasers to specify 50 or 75mm slump concrete is perplexing. Handling, placing and compacting concrete at this consistence is difficult and it is not surprising that anecdotal evidence points to the frequent addition of unauthorised water

***Current Practice Sheet No. 149***

This is the first of a two-part article on Admixture current practice. Part 2 is due to appear in the October 2006 issue of *CONCRETE*.